# XINGCHEN SHA

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## EDUCATION

**Ph.D. in Computer Science** — May 2029 (expected)
Northwestern University, Evanston, United States — Advisor: Edith Elkind

**B.A. in Computer Science** — May 2025
Columbia University, New York, United States — GPA: 4.03

**B.S. in Computer Science** — May 2025
City University of Hong Kong, Kowloon Tong, Hong Kong — GPA: 4.01

## EXPERIENCE

**Input Output Group: Summer Research Intern** — August 2026 – Present

- Researched Strategic Effort in Decentralized Voting
- Advisor: Vangelis Marinakis, Nimrod Talmon

**Columbia University: Research Assistant** — October 2023 – May 2025

- Collaborator in the NSF CSIRO-funded project Fair Sequential Collective Decision Making
- Researched automatic algorithms design with LLMs
- Researched group-fairness in multi-period facility location problems
- Advisor: Haris Aziz, Hau Chan, Toby Walsh, Lirong Xia

**City University of Hong Kong: Research Assistant** — June 2022 – April 2023

- Researched facility location games with capacity constraints
- Researched multi-stage facilty location games in dynamic settings
- Researched truthful mechanism design with optional preferences
- Advisor: Hau Chan, Vincent Chau, Ken Fong, Minming Li, Pinyan Lu, Jialin Zhang

## ACTIVITY

**Columbia University: Student Helper** — September 2024 – May 2025
Decentralized Finance and Game Theory Seminar

**City University of Hong Kong: Spotlight Presenter** — June 2023
STEM Carnival cum Student Project Exhibition 2023

**City University of Hong Kong: Organizer** — May 2023 – September 2023
Hong Kong Secondary School Coding Challenge 2023 (HKSC2023)

**City University of Hong Kong: Student Helper** — September 2022 – September 2023
CityU Theory Seminar

**City University of Hong Kong: Organizer** — May 2022 – September 2022
Hong Kong Secondary School Coding Challenge 2022 (HKSC2022)

## SERVICES

**City University of Hong Kong: College of Engineering Student Chapter Panel** — November 2022 - May 2025
**City University of Hong Kong: Freshman Mentor** — September 2022 - December 2022

## SELECTED HONORS & AWARDS

**College of Computing College Medal** — January 2026
**GS Honor Society** — February 2025
**Upsilon Pi Epsilon** — September 2024

**Dean's List at Columbia University** September 2023 – January 2025
**VTech Group of Companies Scholarship** June 2023
**Dean's List at City University of Hong Kong** September 2020 – May 2023
**The College of Engineering Dean's Scholarship** April 2023
**HKSAR Government Scholarship** February 2023
**CityU Scholarship** February 2022

## PUBLICATIONS (* ALPHABETICAL ORDER )

**Large Language Models for Indivisible Participatory Budgeting**
Nguyen Thach, **Xingchen Sha**, Hau Chan, AAMAS 2026

**Mechanism Design for Facility Location Problems with Capacity Constraints in Bounded Location Space**
**Xingchen Sha**, Hau Chan, Vincent Chau, Ken Fong, Minming Li, Wai Lun Lo, ECAI 2025

**Strategyproof Mechanism for Two Heterogeneous Facilities with Constant Approximation Ratio ***
Minming Li, Pinyan Lu, **Xingchen Sha**, Yuhao Yao, Jialin Zhang, Games and Economic Behavior

**Facility Location Games with Optional Preferences: A Revisit**
**Xingchen Sha**, Shuyu Bao, Hau Chan, Vincent Chau, Ken Fong, Minming Li, AAAI 2025

**Randomized Strategyproof Mechanisms for Multi-stage Facility Location Problem with Capacity Constraints**
Ken Fong, **Xingchen Sha**, Hau Chan, Vincent Chau, Wai Lun Lo, IJTCS-FAW 2024

**Group Fairness in Multi-period Mobile Facility Location Problems ***
Haris Aziz, Hau Chan, **Xingchen Sha**, Toby Walsh, Lirong Xia, AAMAS 2025 (Extended Abstract), ADT 2024 (Abstract)

## SKILLS

**Programming Languages:**
C/C++, Python, Java, SQL, Bash, HTML

**Professional Applications:**
Latex, Matlab, PyTorch, TensorFlow, Git